

useMemo – Use Cases

1. For Expensive Calculations

- Sorting large arrays
- Filtering large lists
- Complex mathematical calculations
- Data transformation
- Prevents recalculating on every render.

2. Memoizing Derived Data

- When creating new objects or arrays from state/props
- Example:
 - { name, age }
 - filteredUsers
 - sortedProducts
- Prevents unnecessary child re-renders.

3. Performance Optimization in Large Apps

- When component re-renders frequently
- When computation is slow
- When working with large datasets

useCallback – Use Cases

1. Passing Functions to Child Components

- When child is wrapped in `React.memo`
- Prevents unnecessary child re-renders

2. Preventing Infinite `useEffect` Loops

- When a function is inside dependency array
- Keeps function reference stable

3. Stable Event Handlers

- Button click handlers
- Form submit handlers
- API call functions

4. Performance Optimization in Re-rendering

- When parent re-renders frequently
- When function is passed down multiple levels